

**RETAIL MARKET PERFORMANCE
ANALYSIS**

PROJECT REPORT

PROBLEM STATEMENT:

In a competitive retail market, businesses need to track sales performance, understand customer behavior, and identify growth opportunities. This project aims to build an end-to-end analytics solution that enables insights generation from raw transactional data using a multi-tool approach.

INTRODUCTION:

Retail Market Performance Analysis is the process of analysing retail data to understand how a business is performing. Retail businesses generate data from customers, orders, products, brands, categories, stores, staff, and inventory. Analysing this data helps to understand sales trends, customer buying behavior, product performance, store performance, and stock availability.

In this project, **Excel, SQL, Power BI, and Python** are used to analyze the retail dataset. SQL is used to extract and combine data from different tables, Excel is used for basic analysis and reporting, Python is used for data cleaning and visualization, and Power BI is used to create an interactive dashboard.

OBJECTIVES:

The main objectives of this project are:

- To analyze retail sales and transaction data effectively.
- To understand customer purchasing behavior and order patterns.
- To monitor product stock and identify inventory-related issues.
- To use SQL for data extraction, transformation, and analysis.
- To use Excel for data exploration, calculations, and summary reports.
- To use Python for data cleaning, exploratory analysis, and visualization.
- To develop an interactive Power BI dashboard for business reporting.
- To identify useful trends and patterns from historical retail data.
- To provide actionable insights that can support better business decisions.

TOOLS & TECHNOLOGIES

The following tools and technologies are used in this project:

- **Excel:** Used for data cleaning, basic calculations, Pivot Tables, charts, and preliminary analysis.
- **SQL:** Used for data extraction, filtering, joining tables, aggregation, and calculating important metrics.
- **Python:** Used for data cleaning, exploratory data analysis, and visualization.
- **Pandas:** Used for data manipulation and analysis.
- **NumPy:** Used for numerical calculations and data processing.
- **Matplotlib:** Used to create charts and visualizations.
- **Seaborn:** Used for statistical and analytical visualizations.
- **Power BI:** Used to create interactive dashboards, KPIs, and business reports.
- **SQL Database:** Used to store and manage the structured retail dataset.

METHODOLOGY:

1. Data Collection

The retail dataset is collected from different tables such as customers, orders, order_items, products, brands, categories, stocks, stores, and staffs. These tables contain information about customers, sales, products, stores, employees, and inventory.

2. Data Cleaning

The collected data is checked for missing values, duplicate records, incorrect values, and data inconsistencies. The data is cleaned to make sure that it is accurate and ready for analysis.

3. SQL Analysis

SQL is used to connect and analyze the different tables. Queries are used to find important information such as total sales, total orders, top products, customer details, and store performance.

4. Excel Analysis

Excel is used for simple data analysis and reporting. Formulas, Pivot Tables, filters, and charts are used to summarize the data and understand sales patterns.

5. Python Analysis

Python is used for further data analysis and visualization. Pandas, NumPy, Matplotlib, and Seaborn are used to clean the data, identify trends, and create graphs.

6. Power BI Dashboard

The analyzed data is imported into Power BI to create an interactive dashboard. The dashboard shows important information such as sales, orders, customers, products, stores, and inventory.

7. Data Analysis and Insights

The results from SQL, Excel, Python, and Power BI are studied to identify important sales trends, customer behavior, product performance, store performance, and inventory conditions.

8. Final Result

Finally, all the findings are summarized to understand the overall retail market performance. The results can help businesses make better decisions and improve their sales and inventory management.

DATASET RELATIONSHIP:

The main relationships between the tables are:

- Customers → Orders → Order Items → Products
- Products → Brands / Categories
- Products → Stocks ← Stores
- Stores → Staffs → Orders

OUTCOMES:

The project provides the following outcomes:

- A clean and organized retail dataset ready for analysis.
- Identification of overall sales and order performance.
- Identification of top-performing products, brands, and categories.
- Analysis of customer purchasing patterns and behavior.
- Comparison of sales performance across different stores.
- Analysis of product demand and inventory availability.
- Identification of low-stock and high-demand products.
- Analysis of staff and order performance.
- SQL-based analysis of important retail metrics.
- Excel-based summaries, calculations, and visualizations.
- Python-based data analysis and trend visualization.
- An interactive Power BI dashboard for monitoring retail performance.
- Useful insights to support data-driven business decisions.

CONCLUSION:

The Retail Market Performance Analysis Using Excel, SQL, Power BI and Python project demonstrates how raw retail data can be converted into meaningful business insights. The project analyzes different aspects of the retail business, including sales, customers, products, brands, categories, stores, staff, and inventory. SQL helps in extracting and analyzing data, Excel supports basic analysis and reporting, Python provides data cleaning and visualization, and Power BI presents the results through an interactive dashboard.